Induction machine fundamentals

Practice questions · Day 10 · Mon 2 Nov · pack M7

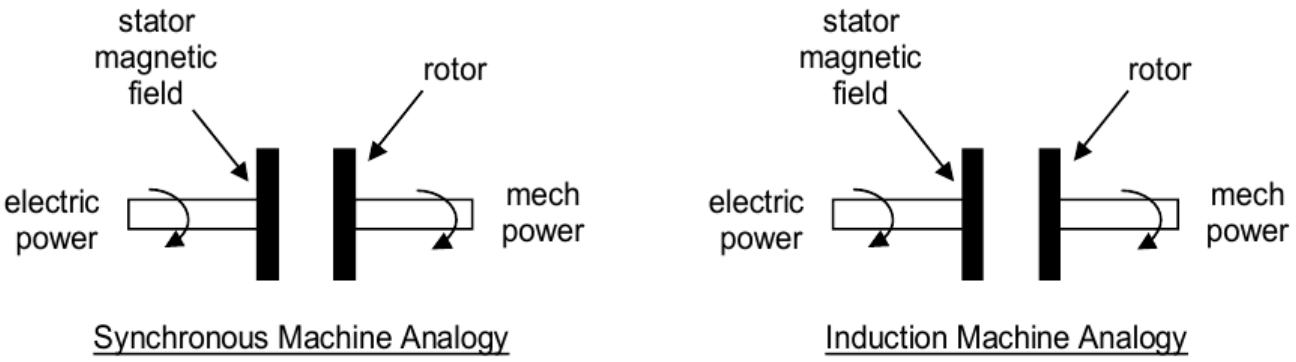
5. Practice

From your course files

PQ 7.1 (practice questions). Answer the following questions about induction machines.

a. The rotor winding of an induction machine consists of _____.
b. List three advantages of induction motors.
c. The most common type of induction motor rotor is called the _____ rotor. The rotor winding in this rotor normally consists of _____ bars.
d. To a first approximation the torque of an induction machine is directly proportional to _____.
e. An induction machine acts a generator when the rotor speed is _____ than the synchronous speed. It acts as a motor when the rotor speed is _____ than the synchronous speed.
f. The equivalent circuit of an induction machine is similar to that of a _____.
g. In an induction machine equivalent circuit, the power conversion between electrical and mechanical power is modelled by a _____.
h. What is the starting and pull-out torque?

PQ 7.2 (practice questions). There are two common mechanical analogies used to describe the operation of synchronous and induction machines. They are both based on modelling the stator magnetic field and the rotor as rotating disks where the stator magnetic field is rotating at synchronous speed. Using the templates below, illustrate these analogies. Label any additional elements added to the figures. No explanation of the analogies are required.



PQ 7.2 templates (from the practice questions): stator-field disk and rotor disk on a shaft, electric power in on the left and mechanical power out on the right, for the synchronous and the induction machine

PQ 7.3 (practice questions). Calculate the synchronous speed (in rpm) for the following AC machines:

a. 4 pole, 50 Hz (b) 8 pole, 60 Hz (c) 2 pole, 50 Hz (d) 4 pole, 800 Hz (e) 24 pole, 50 Hz

PQ 7.4 (practice questions). Calculate the most likely number of poles and supply frequency for motors with the following synchronous speeds.

a. 1,500 rpm (b) 1800 rpm (c) 750 rpm (d) 1,200 rpm (e) 360 rpm (f) 150 rpm

PQ 7.5 (practice questions). Calculate the number of poles, the synchronous speed and the rated slip speed in rpm for three-phase induction motors which have nameplate data as follows:

a. 415 V, 50 Hz, 1440 rpm (b) 3.3 kV, 60 Hz, 3520 rpm (c) 135 V, 50 Hz, 680 rpm (d) 230 V, 50 Hz, 950 rpm

PQ 7.8 (practice questions). Calculate the rated slip and rated slip frequency for the four motors in the question three questions prior to this one.

PQ 7.6 (practice questions). Consider a 7.5 kW, 2910 rpm three-phase 50Hz induction machine operating at rated voltage. Refer to Section 7.2 of your notes if you have difficulty with this.

a. Calculate its synchronous speed in rpm and hence the rated slip speed in rpm.
b. The machine is running at no-load, calculate its operating speed in rpm.
c. At what speed does rated output torque occur at? Calculate the rated output torque.
d. Calculate the rated operating speed as a percentage of the no load operating speed. Is the description of induction machines as almost fixed speed machines valid?
e. Calculate the speed (in rpm) at which the motor operates with half rated output torque.
f. Calculate the output power corresponding to part (e). Express this as a percentage of rated output power. Why is this not exactly half the rated output power?
g. Calculate the operating speed corresponding to a load of 20 Nm.
h. The machine is driven at 3070 rpm. Is the machine acting as a motor or generator? Calculate the output torque.

PQ 7.7 (practice questions). Consider a large induction machine rated at 1470 rpm, 50 Hz operating at rated voltage. Refer to Section 7.2 of your notes if you have difficulty with this.

a. If the machine is delivering half rated output torque, calculate the operating speed in rpm.
b. The machine is driven as a generator with an input torque equal to rated torque. Calculate its operating speed.

PQ 7.9 (practice questions). A 50 Hz induction machine has a rated speed of 960 rpm and a rotor resistance $R_2' = 1$ ohm. For each of the following speeds, calculate the slip, the frequency of the rotor currents and the value of R_2'/s in the equivalent circuit.

a. locked-rotor (that is, at standstill) (b) no-load (c) rated speed (d) operation as a generator at 1050 rpm

PQ 7.10 (practice questions). Answer the following questions about induction machines.

a. An induction machine acts a generator when its rotor speed is _____ than synchronous speed.
b. What is the slip speed?
c. What are the main differences between the equivalent circuit of an induction machine and a transformer?
d. A 50 Hz induction motor is operating at a speed which is 90% of synchronous speed. What is the rotor frequency and slip?
e. The power crossing the airgap between the stator and rotor is called the _____.
f. This power crossing the airgap is less than the power entering the stator because of what losses?
g. The power crossing the airgap is less than the mechanical output power due to what loss?
h. Why are the rotor iron losses often neglected?
i. Briefly explain how the stator resistance R_1 and rotor resistance referred to the primary R_2' can be obtained?
j. What is the relationship between torque and slip at (i) low values of slip? (ii) high values of slip?
k. As far as the rotor resistance R_2' is concerned, what is the difference between a squirrel-cage rotor and a wound-rotor?
l. Changing the rotor resistance, changes the _____ at which maximum torque occurs but does not change its _____.
m. Constant torque loads require motors with high _____ torque.
n. An induction motor is operating at 85% of synchronous speed. What is its ideal efficiency?
o. Why is the power-factor of induction motors low under no-load conditions?

p. The ratio of pullout torque to rated torque for an induction motor is typically about _____.

q. The effective rotor resistance R_2'/s of an induction machine can be thought of as consisting of two separate resistances. Explain the physical significance of these two resistances.

PQ 7.11 (practice questions). A 2.2 kW three-phase induction machine has the following ratings: 2.2 kW, 50 Hz, 1420 rpm, 415 V, 4.9 A, 0.80 efficiency, 0.78 power-factor.

a. Calculate the synchronous speed in rpm
b. Calculate the rated output torque in Nm
c. Calculate the rated input power to the machine: i) Using the voltage, current and power-factor ii) Using the output and efficiency
d. Calculate the ideal efficiency at rated speed. Why is this different to the actual efficiency?

PQ 4.7 (practice questions, only if you skipped M6). A 0.75 kW induction machine is operating from a 110 V, 60 Hz AC power supply. Its synchronous speed is 1200 rpm and its rated speed is 1100 rpm.

a. Calculate the required supply voltage and frequency to give a synchronous speed of 900 rpm.
b. With the conditions in part (a), calculate the speed corresponding to: (i) rated torque (ii) half rated torque

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New questions

N1. [3 marks] A 6-pole, 60 Hz induction motor runs at 1164 rpm. Its referred rotor resistance is $R_2' = 0.45\,\Omega$. Find the slip, the frequency of the rotor currents, and the two parts of R_2'/s.

N2. [3 marks] A 6-pole, 50 Hz motor runs at 970 rpm. Its rotor copper loss is 400 W. Find the airgap power, the mechanical power and the torque.

N3. [5 marks] A 400 V, 50 Hz, 4-pole, **delta-connected** induction motor runs at 1455 rpm. Per phase, $R_1 = 1.2\,\Omega$, $R_2' = 1.0\,\Omega$ and $X_1 + X_2' = 4\,\Omega$. Neglect the magnetising branch.

a. Find the phase current and the line current. [2]
b. Find the airgap power, the torque and the efficiency. [3]

Circuit for N3: delta phase voltage across R1 = 1.2 ohm, j4 ohm and R2′/s, magnetising branch neglected

N4. [10 marks] A 415 V, 50 Hz, 1440 rpm, star-connected, three-phase induction motor has (referred to the stator) $R_1 = 0.8\,\Omega$, $R_2' = 0.6\,\Omega$, $X_1 = X_2' = 1.5\,\Omega$, $X_M = 40\,\Omega$ and $R_C = 500\,\Omega$. It runs at rated voltage and speed.

a. Find the slip and draw the simplified per-phase equivalent circuit with values. [2]
b. Find the rotor current I_2', the input current I_1 (magnitude and phase) and the power factor. [3]
c. Find the input power, the airgap power and the output torque. [3]
d. Find the output power and efficiency, and compare it with the ideal efficiency. Account for the difference with a loss breakdown. [2]

N5. [5 marks] For the motor in N4, the supply voltage sags by 10 % (to 373.5 V line). The load torque stays at 63.59 Nm.

a. What torque would the motor produce at the original slip of 0.04? [1]
b. Estimate the new operating speed using the low-slip approximation $\tau \propto V_1^2 s$. [2]
c. Find the exact new slip from the torque formula, and the new rotor copper loss. Comment. [2]

6. Quick recall check

1. A 4-pole, 50 Hz motor runs at 1470 rpm. What is the rotor frequency?
2. Write R_2'/s as two resistors and state what the power in each one represents.
3. $P_{AG} = 10$ kW at $s = 0.05$. What are the rotor copper loss and the mechanical power?
4. In $\tau = P_{AG}/\omega$, which ω do you use: ω_S or the rotor speed? And for P_M?
5. A 690 V delta-connected motor: what is V_1 in the per-phase circuit?